

SIMILARITY SOLUTIONS FOR REACTIVE SHOCK HYDRODYNAMICS*

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Abstract. It is shown how group-theoretic methods, or similarity methods, can be applied to determine the class of self-similar solutions for a one-dimensional, time-dependent problem in shock hydrodynamics, with a chemical reaction taking place behind the shock. The functional form of all reaction rates depending on the state variables is characterized under which the general system of differential equations and boundary conditions admit self-similar solutions. It is shown that a subclass of these solutions can model certain processes in detonation physics.

1. Introduction. The purpose of this paper is to establish, using the general methods described in Bluman and Cole [1], Ovsjannikov [16], or Harrison and Estabrook [10], the entire class of self-similar solutions to a one-dimensional time-dependent problem in shock hydrodynamics with a chemical reaction taking place. In many physical problems where similarity methods are applied, the forms of specific classes of self-similar solutions are obtained from dimensional analysis or informed guessing on the part of the investigators. However, the general group-theoretic methods described in [1], [16], and [10] guarantee the complete determination of the invariance group, and consequently all self-similar solutions to a problem; as a by-product, we are able to characterize analytically the general form of the reaction rate law under which the entire system of differential equations admits a local, Lie group of transformations and hence similarity solutions.

The physical situation that motivates this study is a hydrodynamic medium in which a chemical reaction occurs. The reaction is initiated by a plane shock wave which is introduced into the medium at time $t = 0$ via a driving piston. For detonation waves, for example, it is experimentally observed that after a time the velocity of the shock becomes constant, and a steady-state condition is reached from the viewpoint of an observer riding on the shock. Analytically, this equilibrium solution was studied first by Chapman [2] and Jouget [11], who assumed that the chemical reaction takes place instantaneously in the shock front. Later, their theory was refined by Zeldovich [27], von Neumann [24], and Doering [7], to include a zone of finite width behind the shock where the chemical reaction occurs. A thorough discussion of these steady-state solutions can be found in Courant and Friedrichs [4] or Fickett and Davis [8].

Our attention in this work is directed toward the so-called initiation problem, that is, the time-dependent problem of describing the flow from the initial time $t = 0$ when the piston impacts, to the time when a steady-state solution takes effect. In recent years there has been much interest in experimentally measuring the flow parameters (shock velocity, particle velocity, pressure, . . .) in this regime, and numerical solutions have been extensively developed (e.g., see [19]). A similarity solution, as developed in this work, not only is of interest itself as an analytic solution, but also can be useful in evaluating the accuracy of Lagrangian hydrodynamics codes.

Self-similar solutions in nonreactive shock hydrodynamics have been studied extensively (principally with reference to blast waves) in planar, cylindrical, and

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spherical geometry. In particular we mention the work by Guderley [19], Taylor [23], Sedov [18], Stanyukovich [20], Ovsjannikov [16], Zeldovich and Raizer [28], and Mishkin and Fujimoto [14], [15]. In the reactive case that we consider in this work, Taylor [22] has investigated spherically expanding, self-similar detonation waves in which the chemical reaction takes place instantaneously across the shock front. In the case of a resolved reaction zone, Sternberg [21] has investigated self-similar solutions for a reaction rate proportional to the energy, and Cowperthwaite [6] has used experimental Lagrangian pressure gage measurements in conjunction with similarity solutions to infer information about energy release rates. One novel part of our calculation is that the reaction rate is treated as a completely arbitrary function of the state variables throughout the analysis and is later determined as a necessary condition for the existence of similarity solutions.

2. Formulation of the model. To model the one-dimensional, time-dependent problem stated in the introduction, we will use a Lagrangian description of the flow, with h denoting the Lagrangian position and t denoting time. Our convention is defined by the equation

$$dx = u dt + \frac{v}{v_0} dh,$$

which relates the Eulerian position x to the Lagrangian position h . The quantities p , u , and v , all functions of t and h , will denote pressure, particle velocity, and specific volume, respectively. The dimensionless quantity λ , also a function of t and h , will denote the progress variable (mass fraction of the product) of an irreversible chemical reaction involving a single reactant and single product. It is assumed that both the reactant and the product satisfy the polytropic equation of state with the same polytropic index γ . That is,

$$(1) \quad E_i = e_i + \frac{p_i v_i}{\gamma - 1}, \quad i = 0, 1,$$

where E_i is the specific internal energy, and the subscripts 0 and 1 refer to the reactant and product, respectively. The quantities e_0 and e_1 are constants related to the heats of formation of the substances. The equation of state of the mixture is assumed to be given by the ideal mixing law

$$(2) \quad E = \lambda E_1 + (1 - \lambda) E_0.$$

Assuming local mechanical equilibrium, (1) and (2) give

$$(3) \quad E = e_0 - \lambda q + \frac{pv}{\gamma - 1},$$

where $q = e_0 - e_1$ is the energy liberated per unit mass in the chemical reaction, i.e., the specific heat of reaction. If we further assume that the flow is adiabatic (in general, heat conduction occurs on a much slower time scale compared to other processes in the problem), then the thermodynamical relation

$$(4) \quad dE = -p dv,$$

along with the fact that

$$dE = \left(\frac{\partial E}{\partial p} \right)_{\lambda, v} dp + \left(\frac{\partial E}{\partial v} \right)_{p, \lambda} dv + \left(\frac{\partial E}{\partial \lambda} \right)_{p, v} d\lambda,$$

yield the conservation of energy equation

$$(5) \quad \frac{\partial p}{\partial t} + \frac{\gamma p}{v} \frac{\partial v}{\partial t} - \frac{\Gamma q}{v} \frac{\partial \lambda}{\partial t} = 0,$$

where $\Gamma = v(\partial p / \partial E)_{v, \lambda}$ is the Grüneisen parameter. For a polytropic equation of state, $\Gamma = \gamma - 1$. Equation (5), coupled with the usual equations for conservation of mass and momentum,

$$(6) \quad \frac{\partial v}{\partial t} - v_0 \frac{\partial u}{\partial h} = 0,$$

$$(7) \quad \frac{\partial u}{\partial t} + v_0 \frac{\partial p}{\partial h} = 0,$$

give the governing hydrodynamical equations for the flow, where v_0 is the specific volume of the reactant. We have neglected the viscous term in (7) since it is generally believed that pressure forces dominate the viscous forces in reactive media. Finally, we assume that the chemical reaction law is given by the differential equation

$$(8) \quad \frac{\partial \lambda}{\partial t} = Q(u, p, v, \lambda),$$

where Q is the reaction rate. For the moment we assume no specific form for Q , only that it depends on the states u , p , v , and λ . We note, however, that although it is mathematically possible to allow Q to depend on the particle velocity u , it is not acceptable as a physical model, since within a Lagrangian framework such a system would not be Galilean invariant.

In addition to the governing differential equations (5) through (8), we also have the Rankine–Hugoniot jump conditions, which must hold across the shock front as the shock discontinuity propagates into the reactant. These act as boundary conditions for the problem, and are given by

$$(9) \quad v_0(D - u_1) = v_1 D,$$

$$(10) \quad p_1 - p_0 = \frac{D u_1}{v_0},$$

$$(11) \quad E_1 - E_0 = \frac{(p_1 + p_0)(v_0 - v_1)}{2}.$$

Here the subscripts 0 and 1 refer to the quantities immediately in front of and immediately behind the shock, respectively, and D is the velocity of the shock wave. Since we assume no chemical reaction in the shock front, we also have the condition

$$(12) \quad \lambda_1 - \lambda_0 = 0.$$

In order to make the jump conditions more tractable for the similarity analysis, we also make the strong shock assumption, which for polytropic gases leads to

$$(13) \quad D = \frac{\gamma + 1}{2} u_1.$$

Hence, the jump conditions (9)–(11) can be rewritten as

$$(14) \quad \frac{v_1}{v_0} = \frac{\gamma - 1}{\gamma + 1},$$

$$(15) \quad p_1 = \frac{\gamma + 1}{2v_0} u_1^2,$$

$$(16) \quad E_1 - E_0 = \frac{1}{\gamma + 1} v_0 p_1.$$

Thus, the mathematical problem is to generate solutions of the partial differential equations (5) through (8) subject to the boundary conditions (14) through (16) and (12) on the shock front, which must also be determined in the problem. We make no a priori assumption concerning the back boundary or piston path. Indeed, such an assumption cannot be made arbitrarily; rather, the similarity solution forces a back boundary condition which must be satisfied in order for such a solution to exist.

We finally make some remarks concerning our assumption of a polytropic gas equation of state. With reference to the area of detonation physics, which is one principal application of our work, it is unfortunate that more reasonable equations of state are not available for both reacted and unreacted explosives over the relevant pressure range, 20–400 kilobars. In this range, the best detonation product equations of state that are available are rather crude fits (using some restricted set of data) to questionable functional forms. The predictive capabilities of these equations of state (BKW, JCZ, etc.) are not spectacular (see, for example, Cowan and Fickett [5]). In fact, some become very unphysical, e.g., triple-valued, for pressures just above the C – J point. Therefore, considering these remarks plus the fact that the polytropic equation of state is the traditional one used by most authors in this area, we shall assume it here.

3. Calculation of the invariance group. The idea of the calculation is to find a one-parameter infinitesimal group of transformations

$$(17) \quad \begin{aligned} \bar{t} &= t + \varepsilon T, & \bar{h} &= h + \varepsilon H, & \bar{u} &= u + \varepsilon U, \\ \bar{p} &= p + \varepsilon P, & \bar{v} &= v + \varepsilon V, & \bar{\lambda} &= \lambda + \varepsilon \Lambda, \end{aligned}$$

where the generators T , H , U , P , V , and Λ are functions of t , h , u , p , v , and λ , under which the system of partial differential equations (5) through (8) and jump conditions (14) through (16) and (12) are invariant. Here, the jump conditions are treated like boundary conditions on the shock path, which also must be invariant under (17). If it is possible to find such a group, then standard techniques [1] permit the number of independent variables in the problem to be reduced by one, thereby allowing the system of partial differential equations to be replaced by a system of ordinary differential equations.

In order to carry out the analysis we introduce a more convenient notation. Let

$$x^1 = t, \quad x^2 = h, \quad u^1 = u, \quad u^2 = p, \quad u^3 = v, \quad u^4 = \lambda,$$

and define the derivatives

$$p_j^i = \frac{\partial u^i}{\partial x^j}, \quad i = 1, \dots, 4; \quad j = 1, 2.$$

A system of differential equations

$$H_n(x^i, u^j, p_i^j) = 0, \quad n = 1, \dots, 4,$$

is said to be constantly conformally invariant under the infinitesimal group (17) if there exist constants α_{nk} ($n, k = 1, \dots, 4$) such that¹

$$(18) \quad \mathcal{L}H_n = \alpha_{nj}H_j, \quad n = 1, \dots, 4,$$

for all smooth surfaces $u^i = u^i(x^j)$, where $\mathcal{L}$ is the Lie derivative in the direction of the extended vector field

$$A = \xi_x^i \frac{\partial}{\partial x^i} + \xi_u^j \frac{\partial}{\partial u^j} + \xi_{p_i^j}^i \frac{\partial}{\partial p_i^j},$$

where

$$\xi_x^1 = T, \quad \xi_x^2 = H, \quad \xi_u^1 = U, \quad \xi_u^2 = P, \quad \xi_u^3 = V, \quad \xi_u^4 = \Lambda,$$

and where

$$\xi_{p_i^j}^i = \frac{\partial \xi_u^i}{\partial x^j} + \frac{\partial \xi_u^i}{\partial u^k} p_j^k - \frac{\partial \xi_x^k}{\partial x^j} p_k^i - \frac{\partial \xi_x^k}{\partial u^m} p_k^i p_j^m$$

are the generators of the derivative transformations. Writing out (18) gives

$$\frac{\partial H_n}{\partial x^i} \xi_x^i + \frac{\partial H_n}{\partial u^i} \xi_u^i + \frac{\partial H_n}{\partial p_j^i} \xi_{p_j^i}^i = \alpha_{nj} H_j, \quad n = 1, \dots, 4.$$

Substituting the $\xi_{p_j^i}^i$ into the last equation gives a polynomial in the p_j^i . Setting the coefficients of p_j^i and $p_j^i p_k^l$ to zero gives a system of first order, linear partial differential equations in the generators $T, H, U, \dots$. In practice this system, which is called the system of determining equations of the group, can be solved to find the group. A similar analysis holds true for the boundary conditions in the problem.

If this program is carried out for the conservation of mass equation (6) we obtain the determining equations

$$(19) \quad T = T(t, h), \quad H = H(t, h),$$

$$(20) \quad \frac{\partial V}{\partial u} + v_0 \frac{\partial T}{\partial h} = \alpha_{12}, \quad \frac{\partial V}{\partial t} - v_0 \frac{\partial U}{\partial h} = -\alpha_{14} Q,$$

$$(21) \quad \frac{\partial V}{\partial p} = \alpha_{13}, \quad \frac{\partial U}{\partial p} = -\alpha_{12}, \quad \frac{\partial U}{\partial \lambda} = 0,$$

$$(22) \quad \frac{\partial V}{\partial v} - \frac{\partial T}{\partial t} = \alpha_{11} + \alpha_{13} \frac{\gamma p}{v},$$

$$(23) \quad \frac{\partial V}{\partial \lambda} = -\alpha_{13} \frac{\Gamma q}{v} + \alpha_{14},$$

$$(24) \quad \frac{\partial H}{\partial h} - \frac{\partial U}{\partial u} = -\alpha_{11}, \quad \frac{\partial H}{\partial t} + v_0 \frac{\partial U}{\partial v} = 0.$$

¹ We assume the summation convention of summing repeated indices.

In the same way, invariance of the momentum equation (7) yields

$$(25) \quad \frac{\partial P}{\partial v} = \frac{\partial P}{\partial \lambda} = 0,$$

$$(26) \quad \frac{\partial U}{\partial u} - \frac{\partial T}{\partial t} = \alpha_{22},$$

$$(27) \quad \frac{\partial U}{\partial p} - v_0 \frac{\partial T}{\partial h} = \alpha_{23}, \quad \frac{\partial H}{\partial t} - v_0 \frac{\partial P}{\partial u} = \alpha_{21} v_0,$$

$$(28) \quad \frac{\partial U}{\partial v} = \alpha_{21} + \alpha_{23} \frac{\gamma p}{v}, \quad \frac{\partial U}{\partial \lambda} = -\alpha_{23} \frac{\Gamma q}{v} + \alpha_{24},$$

$$(29) \quad \frac{\partial P}{\partial p} - \frac{\partial H}{\partial h} = \alpha_{22},$$

$$(30) \quad \frac{\partial U}{\partial t} + v_0 \frac{\partial P}{\partial h} = -\alpha_{24} Q.$$

Next, invariance of the energy equation (5) gives

$$(31) \quad \frac{\partial H}{\partial t} = 0, \quad \alpha_{31} = \alpha_{32} = 0,$$

$$(32) \quad \frac{\partial P}{\partial u} + \frac{\gamma p}{v} \frac{\partial V}{\partial u} - \frac{\Gamma q}{v} \frac{\partial \Lambda}{\partial u} = 0,$$

$$(33) \quad \frac{\partial P}{\partial p} - \frac{\partial T}{\partial t} + \frac{\gamma p}{v} \frac{\partial V}{\partial p} - \frac{\Gamma q}{v} \frac{\partial \Lambda}{\partial p} = \alpha_{33},$$

$$(34) \quad \frac{\partial V}{\partial v} - \frac{\partial T}{\partial t} - \frac{\Gamma q}{\gamma p} \frac{\partial \Lambda}{\partial v} + \frac{P}{p} - \frac{V}{v} = \alpha_{33},$$

$$(35) \quad \frac{\gamma p}{\Gamma q} \frac{\partial V}{\partial \lambda} - \frac{\partial \Lambda}{\partial \lambda} + \frac{\partial T}{\partial t} + \frac{V}{v} = -\alpha_{33} + \alpha_{34} \frac{v}{\Gamma q},$$

$$(36) \quad \frac{\partial P}{\partial t} + \frac{\gamma p}{v} \frac{\partial V}{\partial t} - \frac{\Gamma q}{v} \frac{\partial \Lambda}{\partial t} = -\alpha_{34} Q.$$

Finally, invariance of the species equation (8) gives

$$(37) \quad \frac{\partial \Lambda}{\partial u} = \alpha_{41} = \alpha_{42} = 0, \quad \frac{\partial \Lambda}{\partial p} = \alpha_{43}, \quad \frac{\partial \Lambda}{\partial v} = \alpha_{43} \frac{\gamma p}{v},$$

$$(38) \quad \frac{\partial \Lambda}{\partial \lambda} - \frac{\partial T}{\partial t} = \alpha_{44} - \alpha_{43} \frac{\Gamma q}{v},$$

$$(39) \quad \frac{\partial \Lambda}{\partial t} = U \frac{\partial Q}{\partial u} + P \frac{\partial Q}{\partial p} + V \frac{\partial Q}{\partial v} + \Lambda \frac{\partial Q}{\partial \lambda} - \alpha_{44} Q.$$

The solution of these determining differential equations for the group is not difficult. From (37) we immediately obtain $\alpha_{43} = 0$, from which it follows that Λ is independent of p and v . Equations (27) and (28) give $\alpha_{23} = 0$; hence $\alpha_{24} = 0$, $\alpha_{21} = 0$, and $\partial P / \partial u = 0$. Then (32) gives $\partial V / \partial u = 0$. Next, (22) and (23) give $\alpha_{13} = 0$, which in turn yields $\partial V / \partial p = 0$. Combining (20), (21), and (27) gives $\alpha_{12} = 0$, which leads to $\partial U / \partial p = \partial T / \partial h = 0$. Next, from (35) it follows that $\partial V / \partial \lambda = 0$ and so $\alpha_{14} = 0$. Finally $\alpha_{34} = 0$ follows from (36). In summary, we make the following formal remark.

Remark. Since $\alpha_{ij} = 0$ for $i \neq j$, the general invariance condition (18) reduces to the specialized condition $\mathcal{L}H_n = \alpha_{nn}H_n$, $n = 1, \dots, 4$ (no sum on n).

It is clear from the determining equations (19)–(39) that

$$\frac{\partial^2 V}{\partial v^2} = \frac{\partial^2 U}{\partial u^2} = \frac{\partial^2 \Lambda}{\partial \lambda^2} = \frac{\partial^2 P}{\partial p^2} = \frac{\partial^2 T}{\partial t^2} = 0.$$

Hence $T = at + c$, from which we instantly conclude that $U = (\alpha_{22} + a)u + f(t, h)$, $P = (\alpha_{33} + a)p + g(t, h)$, $V = (\alpha_{11} + a)v + w(t, h)$, $\Lambda = (\alpha_{44} + a)\lambda + y(t, h)$, and $H = (\alpha_{22} - \alpha_{11} + a)h + d$. Equation (34) gives $g(t, h) = w(t, h) = 0$, from which we obtain $\partial \Lambda / \partial t = \partial U / \partial t = \partial U / \partial h = 0$. Finally, (29) and (35) give the relations

$$(40) \quad \alpha_{33} + \alpha_{11} - 2\alpha_{22} = 0,$$

and

$$(41) \quad \alpha_{11} + a - \alpha_{44} = -\alpha_{33}.$$

Therefore, if we define

$$(42) \quad b = \alpha_{22} - \alpha_{11} + a,$$

then the generators of the invariance group for the differential equations (5) through (8) are given by

$$(43) \quad \begin{aligned} T &= at + c, & H &= bh + d, \\ U &= (b + \alpha_{11})u + f, & p &= (a + \alpha_{33})p, \\ V &= (a + \alpha_{11})v, & \Lambda &= (a + \alpha_{44})\lambda + y(h), \end{aligned}$$

where a, b, c, d , and f are constants and $y(h)$ is an arbitrary function.

The requirement that the jump conditions also remain invariant under the group places further restrictions on the group. Invariance of (14) easily gives $\alpha_{11} = -a$, and (15) gives $f = 0$. Next, (12) yields $y(h) = 0$, while (16) leads to $\alpha_{44} = \alpha_{33}$. Therefore, we can summarize our main result as follows:

THEOREM 1. *Under the polytropic gas and strong shock assumptions, the equations for reactive flow, equations (5)–(8), and the Rankine–Hugoniot jump conditions (14)–(16) and (12), are constantly conformally invariant under the infinitesimal group (17) with generators given by*

$$(44) \quad \begin{aligned} T &= at + c, & H &= bh + d, & U &= (b - a)u, \\ P &= 2(b - a)p, & V &= 0, & \Lambda &= 2(b - a)\lambda; \end{aligned}$$

here the reaction rate $Q = Q(u, p, v, \lambda)$ must satisfy the partial differential equation

$$(45) \quad \frac{u}{2} \frac{\partial Q}{\partial u} + p \frac{\partial Q}{\partial p} + \lambda \frac{\partial Q}{\partial \lambda} = \beta Q,$$

where

$$(46) \quad \beta = \frac{2b - 3a}{2b - 2a}.$$

The partial differential equation (45) follows immediately from (39). It can be integrated by the method of characteristics to find the general form of the reaction rate

Q . The characteristic system is

$$\frac{du}{u/2} = \frac{dp}{p} = \frac{dv}{0} = \frac{d\lambda}{\lambda} = \frac{dQ}{\beta Q},$$

and has first integrals

$$v, \quad \frac{u^2}{p}, \quad \frac{\lambda}{p}, \quad \text{and} \quad \frac{Q}{p^\beta}.$$

Consequently, the general solution of (45) can be written

$$(47) \quad Q = p^\beta F\left(v, \frac{\lambda}{p}, \frac{u^2}{p}\right),$$

where F is an arbitrary function. Therefore, we have the following theorem.

THEOREM 2. *Under the polytropic gas and strong shock assumptions, a necessary condition for the equations of reactive flow and jump conditions to admit the local, Lie group with generators given by (44) is that the reaction rate Q have the form*

$$Q = k\left(\frac{p}{p_i}\right)^\beta F\left(\frac{v}{v_i}, \frac{p_i\lambda}{p}, \frac{p_i u^2}{p u_i^2}\right),$$

where F is arbitrary, where p_i , v_i , and u_i are the initial ($t = 0$) values of pressure, volume, and particle velocity behind the shock, and where k is a constant having units 1/time.

4. Construction of self-similar solutions. Once the infinitesimal generators of the local Lie group of transformations are known, the invariant surface condition [1, p. 150] can be used to determine the similarity variable and the similarity solution. In the present case, this condition becomes, for u , p , v , and λ respectively,

$$(48) \quad T \frac{\partial u}{\partial t} + H \frac{\partial u}{\partial h} = U,$$

$$(49) \quad T \frac{\partial p}{\partial t} + H \frac{\partial p}{\partial h} = P,$$

$$(50) \quad T \frac{\partial v}{\partial t} + H \frac{\partial v}{\partial h} = V,$$

$$(51) \quad T \frac{\partial \lambda}{\partial t} + H \frac{\partial \lambda}{\partial h} = \Lambda.$$

The characteristic system of (48) is

$$\frac{dt}{at+c} = \frac{dh}{bh+d} = \frac{du}{(b-a)u}.$$

One first integral gives the *similarity variable*

$$(52) \quad s = \frac{c_4 h + 1}{(c_3 t + 1)^{c_2}},$$

where

$$(53) \quad c_2 = \frac{b}{a}, \quad c_3 = \frac{a}{c}, \quad \text{and} \quad c_4 = \frac{b}{d};$$

the second first integral gives

$$(54) \quad u(t, h) = (c_3 t + 1)^{c_2 - 1} u_i \hat{u}(s),$$

where $\hat{u}(s)$ is a function of s and u_i is the initial particle velocity behind the shock. In the same manner, (49), (50), and (51) lead to

$$(55) \quad p(t, h) = (c_3 t + 1)^{2(c_2 - 1)} p_i \hat{p}(s),$$

$$(56) \quad v(t, h) = v_i \hat{v}(s),$$

$$(57) \quad \lambda(t, h) = (c_3 t + 1)^{2(c_2 - 1)} \hat{\lambda}(s).$$

Here p_i and v_i are the initial values of p and v behind the shock; they are given from the jump conditions by

$$v_i = \frac{\gamma - 1}{\gamma + 1} v_0, \quad p_i = \frac{\gamma + 1}{2v_0} u_i^2.$$

Since the shock must be a similarity curve and pass through $t = 0$, $h = 0$, it follows that the shock path is given by

$$(58) \quad c_4 h + 1 = (c_3 t + 1)^{c_2}.$$

Hence, the *shock velocity* is

$$(59) \quad D = D_0 (c_3 t + 1)^{c_2 - 1},$$

where

$$(60) \quad D_0 = \frac{c_2 c_3}{c_4} = \frac{\gamma + 1}{2} u_i$$

is the initial shock velocity.

By substituting the self-similar forms of the solution, (54) through (57), into the differential equations (5) through (8), we obtain a system of ordinary differential equations in $\hat{u}$, $\hat{p}$, $\hat{v}$, and $\hat{\lambda}$, with the similarity variable s acting as the independent variable. Under this calculation, the mass, momentum, energy, and species equations become, respectively,

$$(61) \quad \hat{u}' + \frac{\gamma - 1}{2} s \hat{v}' = 0,$$

$$(62) \quad s \hat{u}' - \hat{p}' = \frac{\hat{u}}{3 - 2\beta},$$

$$(63) \quad s \hat{p}' + \frac{\gamma s \hat{p}}{\hat{v}} \hat{v}' = \frac{2\hat{p}}{3 - 2\beta} - \frac{4kq\hat{p}^\beta F}{(\gamma + 1)c_4 u_i^3 \hat{v}},$$

$$(64) \quad s \hat{\lambda}' = \frac{2\hat{\lambda}}{3 - 2\beta} - \frac{2k\hat{p}^\beta F}{(\gamma + 1)c_4 u_i},$$

where $F = F(\hat{v}, \hat{\lambda}/\hat{p}, \hat{u}^2/\hat{p})$ is an arbitrary function. The initial conditions are given at $s = 1$ by

$$(65) \quad \hat{u}(1) = \hat{p}(1) = \hat{v}(1) = 1, \quad \hat{\lambda}(1) = 0.$$

In summary, then, the mathematical problem of determining self-similar solutions has been reduced to solving the system of ordinary differential equations (61)–(64) subject to the initial conditions (65). Clearly we have $1 < s < \infty$.

The results obtained above by the general method of local Lie groups could have been obtained in a more efficient manner by the classical method of scaling (see, for example, Sedov [18]) where one attempts to find solutions of the conventional power law type, $p = t^m \hat{p}(ht^{-n})$, etc. The main thrust of the present work is to show that these solutions are the *only* similarity solutions; the group-theoretic method used here guarantees the entire class of solutions.

5. Discussion of the solution. In this section we discuss the qualitative features of the various possible solutions implied by the form of the similarity variable s and the choice of the constants in the problem. Of the various constants which occur in the differential equations (61)–(64), only one is free, namely c_4 , a number having units of inverse length representing a characteristic length (or length scale) for the problem. The remaining quantities in the problem are assumed to be known either from the form of the rate law (k , β , and F), the physical characteristics of the reactive material (γ and q), or the initial piston energy (u_i). Actually, the rate multiplier k , whose inverse represents a characteristic time for the problem can be scaled out of the problem originally.

The constant c_2 occurring in the expression for the similarity variable is related to the pressure exponent β via

$$c_2 = \frac{2\beta - 3}{2\beta - 2}.$$

This equation clearly distinguishes three cases:

$$\text{I. } \beta > \frac{3}{2} \quad \text{II. } 1 < \beta < \frac{3}{2} \quad \text{III. } \beta < 1.$$

In each case the constant c_4 can be positive or negative; hence, six possibilities arise. In case I with $c_4 < 0$ the shock is accelerating, and the pressure p_s at the shock, which is given by

$$(66) \quad p_s(t) = p_i(c_3 t + 1)^{2(c_2 - 1)},$$

is increasing. In this case the similarity curves $s = \text{constant}$ meet at the point $(-1/c_4, -1/c_3)$ (see Fig. 1). Case II with $c_4 > 0$ again gives rise to an accelerating shock

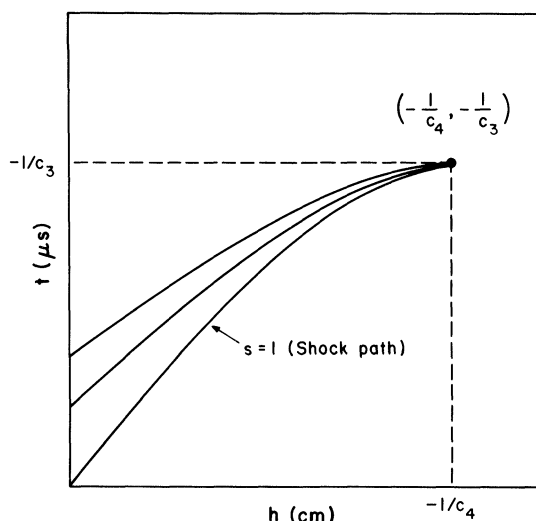


FIG. 1. Similarity curves $s = \text{constant}$ for the case $\beta > 3/2$, $c_3 < 0$, and $c_4 < 0$, where s is given by (52). The curve $s = 1$ is the shock trajectory. The other curves shown are for $s > 1$.

and increasing pressure p_s ; however, in this case the similarity curves are asymptotic to the line $t = -1/c_3$. In case III with $c_4 > 0$ the shock is again accelerating and p_s is increasing; the similarity curves fan out. The remaining three cases, case I with $c_4 > 0$, case II with $c_4 < 0$, and case III with $c_4 < 0$, all represent decelerating shocks with p_s decreasing at the shock front.

It is also informative to compute the derivatives of particle velocity, pressure, volume, and $\hat{\lambda}$ at the shock front. From (64) we conclude that

$$\hat{\lambda}'(1) = -\frac{2kf}{(\gamma+1)c_4u_i},$$

where $f = F(1, 0, 1) > 0$. Hence we must eliminate the case $c_4 > 0$ from consideration since in that case we would have $\lambda < 0$ behind the shock front. Therefore, to model reactive processes with accelerating shocks, only case I with $c_4 < 0$ is relevant. The gradients of the particle velocity, pressure, and specific volume at the shock are easily found from (61)–(63) to be

$$\begin{aligned}\hat{u}'(1) &= \frac{1-\gamma}{1+\gamma} \left[\frac{3}{3-2\beta} - \frac{4kqf}{(\gamma+1)c_4u_i^3} \right], \\ \hat{p}'(1) &= \frac{2}{\gamma+1} \left[\frac{1-2\gamma}{3-2\beta} + \frac{2(\gamma-1)kqf}{(\gamma+1)c_4u_i^3} \right], \\ \hat{v}'(1) &= \frac{2}{1+\gamma} \left[\frac{3}{3-2\beta} - \frac{4kqf}{(\gamma+1)c_4u_i^3} \right].\end{aligned}$$

A high pressure dependence, i.e., a large value of β , in the reaction rate will obviously have the effect of driving the pressure down behind the shock front. Also, increasing the factor q/u_i^2 , which is the ratio of energy of the chemical reaction to the energy induced by the piston, produces a similar effect.

Also, we note that the steady-state solution is a special case of our general results. Choosing $a = b = 0$, $c = 1$, and $d = D$ (the steady-state detonation velocity) in (44) we obtain the similarity variable

$$s = h - Dt,$$

which represents a family of parallel straight lines. Again, $s = 1$ gives the shock path, and p , u , v , and λ are constant along the similarity lines. In this special case, (45) is satisfied identically for *any* reaction rate Q , and therefore, consistent with the known theory, in the steady-state case a similarity solution exists for any reaction rate.

Finally, we note that the case $\beta = 1$ represents a special case. In the limit as $\beta \rightarrow 1$ ($|c_2| \rightarrow +\infty$), the similarity variable s becomes

$$s = (c_4t + 1) \exp \left(-\frac{\gamma+1}{2} u_i c_4 t \right),$$

while the self-similar forms of the state variables are

$$\begin{aligned}u &= u_i \hat{u}(s) \exp \left(\frac{\gamma+1}{2} u_i c_4 t \right), \\ p &= p_i \hat{p}(s) \exp ((\gamma+1) u_i c_4 t), \\ v &= v_i \hat{v}(s), \\ \lambda &= \hat{\lambda}(s) \exp ((\gamma+1) u_i c_4 t).\end{aligned}$$

The same governing differential equations (61)–(64) hold true with $\beta = 1$.

6. Application to detonation. The class of similarity solutions (case I with $c_4 < 0$) derived in the last section can reproduce some of the essential features of experimentally observed flows involving solid reactants. The choice of the free constant c_4 and the constant c_3 can be made from existing “pop-plot” data (see [17]) which relate the input pressure p_i to the distance $\bar{h}$ and time $\bar{t}$ that the shock must run before developing into a full detonation. That is, we select

$$c_3 = -1/\bar{t}, \quad c_4 = -1/\bar{h}.$$

Therefore, the point $(-1/c_4, -1/c_3)$ [see Fig. 1] is the transition-to-detonation point in our model. These choices then force values for c_2 and β via (60). The choice of the function F in the rate law is arbitrary. In general, little is known concerning the form of the rate law. A p^β factor, however, is not inconsistent with available data on flame propagation in the burning of propellants at high pressures; moreover, the presence of a p^β factor is in agreement with the pressure-controlled rate laws used by Mader and Forest [13] and Cochran [3] in numerical calculations. A depletion factor which cuts off the reaction can be justified as well. Therefore, we assume that the reaction rate is given by

$$Q(p, \lambda) = k \left(\frac{p}{p_i} \right)^\beta \left(1 - \frac{c_1 p_i \lambda}{p} \right),$$

which is consistent with the requirements of the similarity solution. The presence of an additional constant c_1 can aid in calibrating the calculated solution with experimental results.

We choose a numerical example: in the explosive PBX-9404, with specific volume $v_0 = 0.5291 \text{ cm}^3/\text{g}$ and heat of detonation $q = 1.38 \text{ Kcal/g}$, at $p_i = 2.95 \text{ GPa}$ pop-plot data gives $\bar{h} = 0.994 \text{ cm}$ and $\bar{t} = 2.46 \text{ } \mu\text{s}$. Hence $c_4 = -1.006 \text{ cm}^{-1}$ and $c_3 = -0.4065 \text{ } \mu\text{s}^{-1}$. If $\gamma = 7$, it follows that $D_0 = 0.2499 \text{ cm}/\mu\text{s}$ and so $c_2 = 0.6184$ and $\beta = 2.3102$. We chose the constants k and c_1 to be $k = -c_4 u_i = 0.0629 \text{ } \mu\text{s}^{-1}$ and $c_1 = 55.0$, in order to release the chemical energy over a time-scale of interest. Using a fourth-order Runge-Kutta routine, the system of equations (61)–(64) was integrated numerically. Fig. 2 shows the computed shock velocity D , and compares it with experimental values taken at the same input pressure p_i . Figs. 3, 4, and 5 show the calculated time histories of particle velocity, pressure, and volume at the Lagrangian points $h = 0.1, 0.3, 0.5$, and 0.8 cm . Fig. 6 shows the chemical progress variable λ at the Lagrangian positions $h = 0.1$ and 0.8 cm . The pressure histories are in qualitative agreement with data taken by several investigators (e.g., Kanel’ and Dremin [12], Wackerle [15], and Weingart et al. [26]), using Lagrangian pressure gages. They show for a variety of reactive media that compression waves behind the shock tend to grow and overtake the shock front. This type of behavior is clearly characteristic of the similarity solution where the similarity lines converge to the point $(-1/c_4, -1/c_3)$. However, near this point the solutions become singular, and we must discount them in this transition region between the initiation regime and the steady-state solution.

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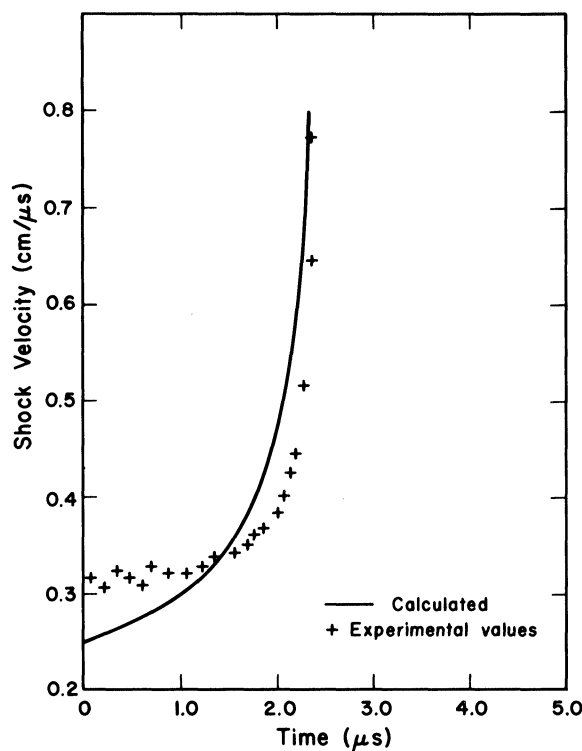


FIG. 2. Shock velocity versus time compared to experimental values at input pressure 2.95 GPa.

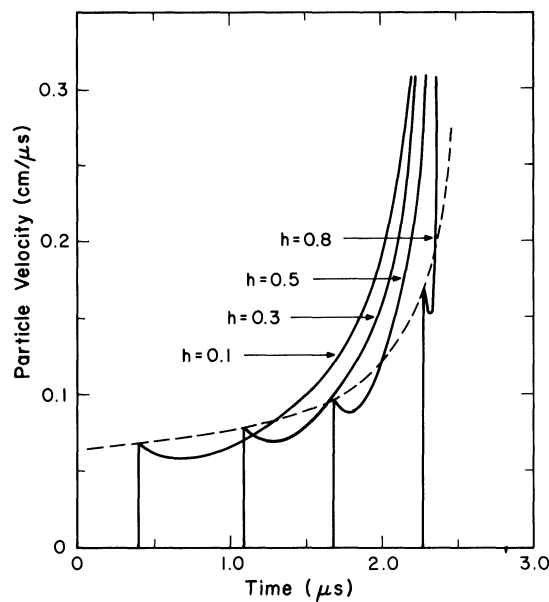


FIG. 3. Particle velocity histories at the Lagrangian positions $h = 0.1, 0.3, 0.5$ and 0.8 cm. The dotted line shows the particle velocity at the shock front.

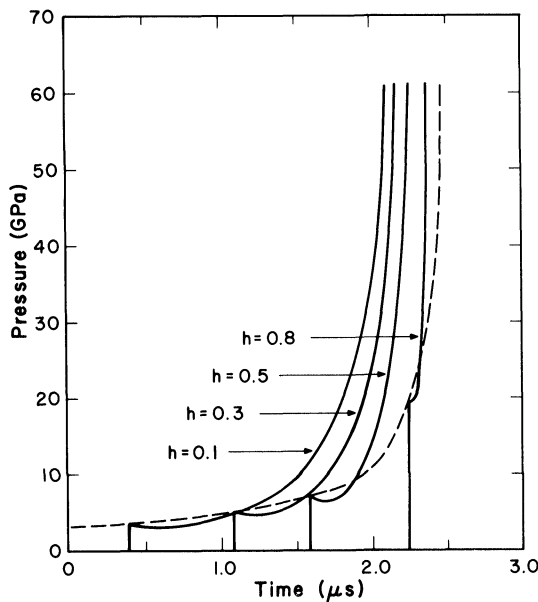


FIG. 4. Pressure histories at the Lagrangian positions $h = 0.1, 0.3, 0.5$ and 0.8 cm. The dotted line shows the pressure buildup at the shock front.

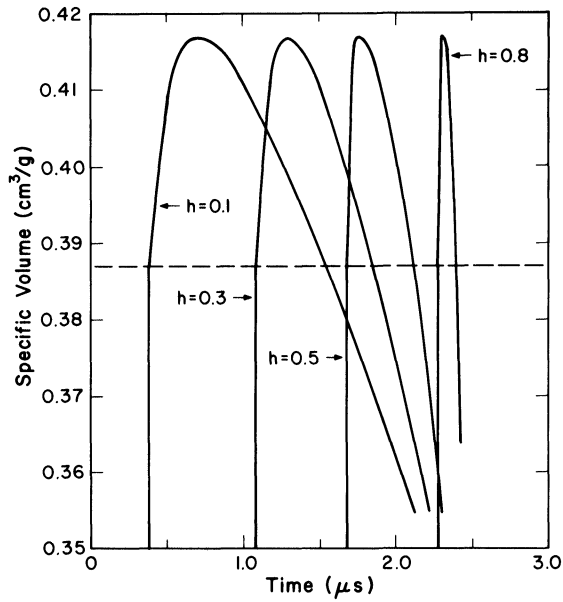


FIG. 5. Specific volume versus time at the Lagrangian positions $h = 0.1, 0.3, 0.5$ and 0.8 cm. The volume at the shock front, shown by the dotted line, is constant.

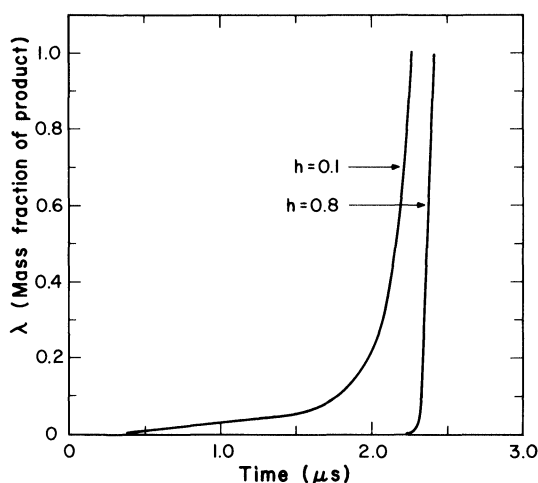


FIG. 6. The chemical progress variable λ versus time at the Lagrangian positions $h = 0.1$ and 0.8 cm.

REFERENCES

- [1] G. W. BLUMAN AND J. D. COLE, *Similarity Methods for Differential Equations*, Springer-Verlag, New York, 1974.
- [2] D. L. CHAPMAN, *On the rate of explosion in gases*. Phil. Mag. and J. Science, 47, Ser. 5 (1899), pp. 90–104.
- [3] S. COCHRAN, L. ERICKSON, R. LEE AND R. WEINGART, *Shock Initiation of TATB*, Seventeenth Symposium on Combustion, The Combustion Institute, 1979.
- [4] R. COURANT AND K. O. FRIEDRICH, *Supersonic Flow and Shock Waves*, Interscience, New York, 1948.
- [5] R. D. COWAN AND W. FICKETT, *Calculation of the detonation properties of solid explosives with the Kistiakowsky–Wilson equation of state*. J. Chem. Phys., 24 (1956), pp. 932–939.
- [6] M. COWPERTHWAIT, *Model solutions for the initiation of condensed explosives*, in Symposium on High Dynamic Pressures, Paris, 1979.
- [7] W. DOERING, *Beitrage zur Theorie der Detonation*, Ann. Physik, 43 (1943), pp. 421–428.
- [8] W. FICKETT AND W. C. DAVIS, *Detonation*, University of California Press, Berkeley, 1979.
- [9] G. GUDERLEY, *Starke Kugelige und zylindrische Verdichtungsstosse in der Nahe des Kugelmittelpunktes bzw. der Zylinderachse*, Luftfahrt-Forschung, 19 (1942), pp. 302–312.
- [10] B. K. HARRISON AND R. B. ESTABROOK, *Geometric approach to invariance groups and solution to partial differential systems*, J. Math. Phys., 12 (1971), pp. 653–666.
- [11] M. E. JOUGET, *Sur la propagation des discontinuités dans les fluides*, Compt. Rend., 132 (1901), pp. 673–676.
- [12] G. KANEL' AND A. DREMIN, *Decomposition of Cast Trotyl in Shock Waves*, Fizika Goreniya i Vzryva, 13 (1977), pp. 85–92.
- [13] C. L. MADER AND C. A. FOREST, *Two Dimensional Homogeneous and Heterogeneous Detonation Wave Propagation*, Report LA-6259, Los Alamos Scientific Laboratory, 1976.
- [14] E. A. MISHKIN AND Y. FUJIMOTO, *Analysis of a cylindrical imploding shock wave*, J. Fluid Mech., 89 (1978), pp. 61–78.
- [15] ———, *Analysis of spherically imploding shocks*, Phys. Fluids, 21 (1978), pp. 1933–1938.
- [16] L. V. OVSIANNIKOV, *Group Properties of Differential Equations*, translation by G. W. Bluman of Gruppovye Svoystva Differentsialny Uravneni, Novosibirsk, U.S.S.R., 1962.
- [17] J. B. RAMSAY AND A. POPOLATO, *Analysis of shock wave and initiation data for solid explosives*, in Fourth Symposium on Detonation, Office of Naval Research, Washington, 1965, pp. 233–238.
- [18] L. I. SEDOV, *Similarity and Dimensional Methods in Mechanics*, Academic Press, New York, 1959.
- [19] *Sixth (International) Symposium on Detonation*, Office of Naval Research, Washington, 1976.
- [20] K. P. STANYUKOVICH, *Unsteady Motion of Continuous Media*, Pergamon Press, New York, 1960.
- [21] H. M. STERNBERG, *Similarity solutions for reactive shock waves*, Quart. J. Mech. and Appl. Math., 23 (1970), pp. 77–99.

- [22] G. I. TAYLOR, *The dynamics of the combustion products behind plane and spherical detonation fronts in explosives*, Proc. Royal Society, 200 (1950), pp. 235–247.
- [23] ———, *The air wave surrounding an expanding sphere*, Proc. Royal Society London A, 186 (1946), pp. 273–292.
- [24] J. VON NEUMANN, *Theory of Detonation Waves*, OSRD Report 549, 1942.
- [25] J. WACKERLE, *Shock initiation of high-density PETN*, in Sixth Symposium on Detonation, Office of Naval Research, Washington, 1976, pp. 20–28.
- [26] D. WEINGART, R. BARLETT, S. COCHRAN, L. ERICKSON, J. CHAN, J. JANSEN, R. LEE, D. LOGAN AND J. ROSENBERG, *Manganin stress gages in reacting high explosives environment*, Symposium on High Dynamic Pressure, Paris, 1979.
- [27] YA. B. ZEL'DOVICH, *On the theory of the propagation of detonation in gaseous systems*, J. of Experimental and Theoretical Physics of the U.S.S.R., 10 (1940), pp. 542–568.
- [28] YA. B. ZEL'DOVICH AND YR. P. RAIZER, *Physics of Shock Waves and High-Temperature Hydrodynamic Phenomena*, Vol. II, Academic Press, New York, 1967.